

Sort the following array using QuickSort

Ex
i < 6

n-1	n-1-i
i	j
0	0 1 2 3 4 5
1	0 1 2 3 4
2	0 1 2 3
3	0 1 2
4	0 1
5	0

first pass (or) pass 0

i=0

j=0, 1, 2, 3, 4, 5

12	54	19	3	8	46	31
----	----	----	---	---	----	----

j=0

Compare

12	19	54	3	8	46	31
----	----	----	---	---	----	----

j=1

Compare

j=2	12	19	3	54	8	46	31
-----	----	----	---	----	---	----	----

Compare

j=3	12	19	3	8	54	46	31
-----	----	----	---	---	----	----	----

Compare

j=4	12	19	3	8	46	54	31
-----	----	----	---	---	----	----	----

Compare

j=5	12	19	3	8	46	31	54
-----	----	----	---	---	----	----	----

fixed element

pass 1 (or) Second pass

i=1 j=0, 1, 2, 3, 4

j=0	12	19	3	8	46	31	54
-----	----	----	---	---	----	----	----

j=0

12	19	3	8	46	31	54
----	----	---	---	----	----	----

Compare

$j=1$

12	3	19	8	46	31	54
----	---	----	---	----	----	----

Compare

$j=2$

12	3	8	19	46	31	54
----	---	---	----	----	----	----

Compare

$j=3$

12	3	8	19	46	31	54
----	---	---	----	----	----	----

Compare

$j=4$

12	3	8	19	31	46	54
----	---	---	----	----	----	----

fixed

pass 2

(or) Third pass

$i=1$

$j=0, 1, 2, 3,$

$j=0$

3	12	8	19	31	46	54
---	----	---	----	----	----	----

Compare

$j=1$

3	8	12	19	31	46	54
---	---	----	----	----	----	----

Compare

$j=2$

3	8	12	19	31	46	54
---	---	----	----	----	----	----

Compare

$j=3$

3	8	12	19	31	46	54
---	---	----	----	----	----	----

fixed

pass 3

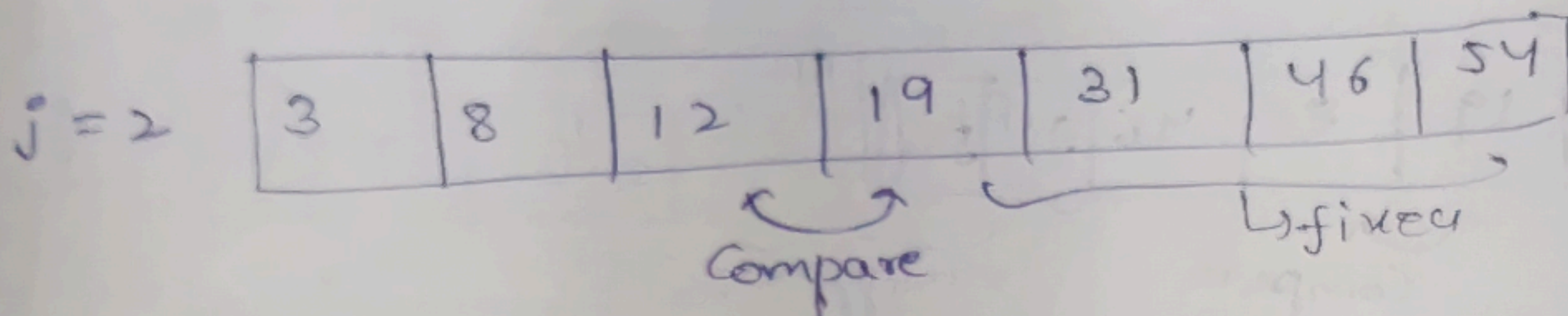
(or) fourth pass

$i=2$

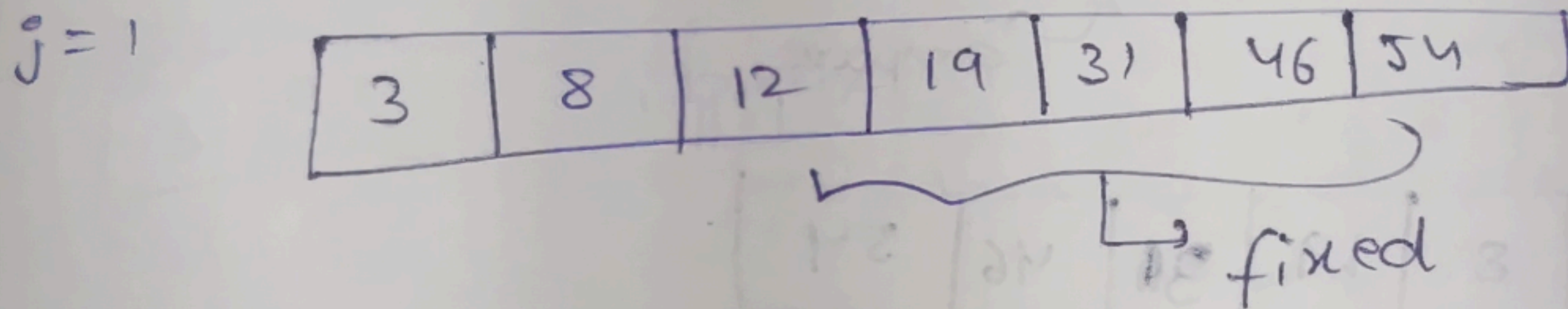
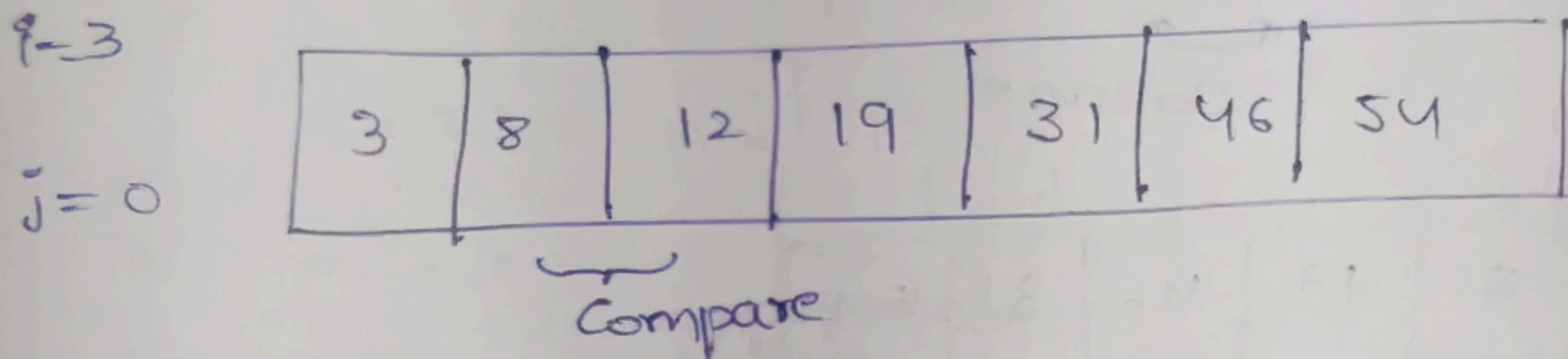
$j=0$

3	8	12	19	31	46	54
---	---	----	----	----	----	----

Compare

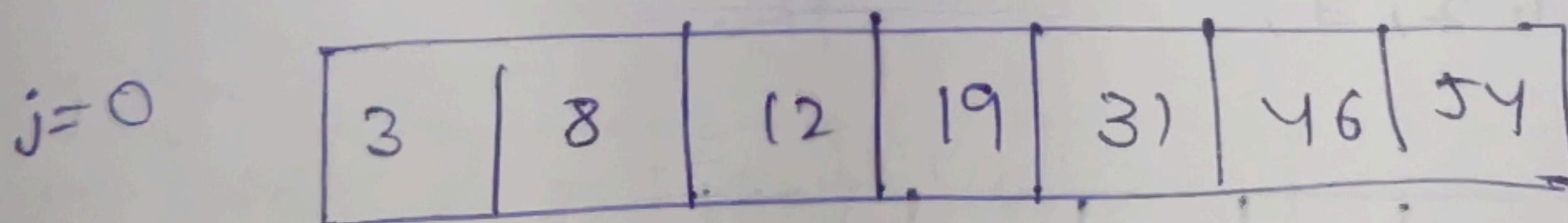


pass 4 (or) fifth pass



pass 5 (or) Sixth pass

$i=4$



~~all are fixed~~, all are fixed

final sorted array

